

PROJECT Report

Digital Logic Design

Instructor

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**Submitted by
Group # 19**

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1. Project Title

Design and Implementation of an Automated **Water Level Monitoring System** using NPN Transistor Switching and 3-Input AND Logic.

2. Introduction / Background

Water wastage due to tank overflow is a common issue in residential and industrial sectors. Manual monitoring is often unreliable and leads to resource depletion or pump damage. This project introduces a digital solution that uses sensors to detect water levels and provides an automated audio-visual alert(alarm) when the tank reaches its maximum capacity.

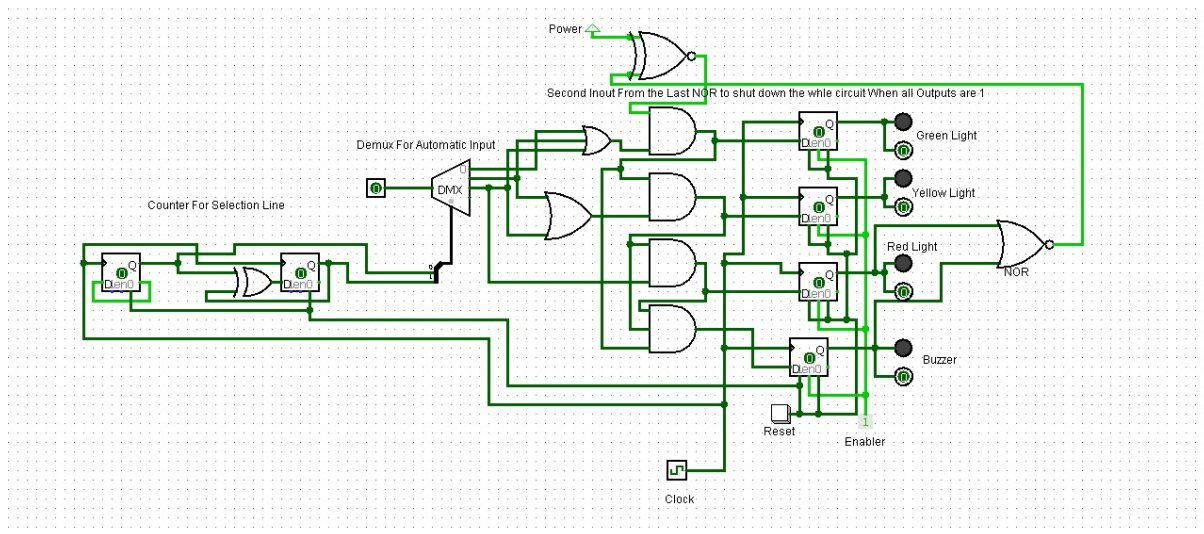
3. Objectives

- To design a low-cost, reliable water level sensor using **BC547 NPN transistors**.
- To implement digital logic using the **74HC11 (Triple 3-Input AND gate)** for decision making.
- To provide visual indication for **three levels (Low, Medium, High)** and a buzzer alarm for the overflow state.

4. Team Members & Contribution

Name	Contribution
Abdul Rafay	Circuit architecture, NPN switching logic, and Tinkercad simulation.
Zain Abbas	Logic Diagram design in Logisim, Proposal and Report writing and Video Editing.
Daim Hassan	K-Maps, Derivation of Truth Tables, PPT and, Block Diagram creation.

5. Logic diagram:



Circuit Overview:

This is an automated sequential control system (likely simulating a safety alarm or traffic/indicator system). It takes a single input, distributes it sequentially using a counter and a demultiplexer, stores the states in D flip-flops, and triggers four outputs (Green Light, Yellow Light, Red Light, and Buzzer). It also features an automatic shutdown mechanism when specific conditions are met.

Key Components & Their Functions

i. The Counter (Selection Line)

What it is: Located on the far left, consisting of two D flip-flops and an XOR gate.

Function: It acts as a 2-bit binary counter that automatically increments on every clock pulse. Its 2-bit output serves as the select line for the Demultiplexer (DMX).

ii. Demultiplexer (DMX for Automatic Input)

What it is: The component labeled DMX in the middle left.

Function: It receives a constant high/data input (the green 1 square). Based on the 2-bit selection lines coming from the counter, it routes this input sequentially to one of its four output channels (Channels 0, 1, 2, and 3).

iii. Combinational Logic (OR & AND Gates)

Function: The outputs of the DMX pass through OR gates and a bank of 4 AND gates. This logic determines the conditions under which each output flip-flop is allowed to turn on.

iv. Storage (D Flip-Flops)

Function: There are 4 D flip-flops in a vertical row on the right. They capture and hold the state for the four main outputs on every clock tick, as long as the system is active (Enabler is high).

v. Outputs

The system drives four indicators based on the state of the flip-flops:

- Green Light
- Yellow Light
- Red Light
- Buzzer

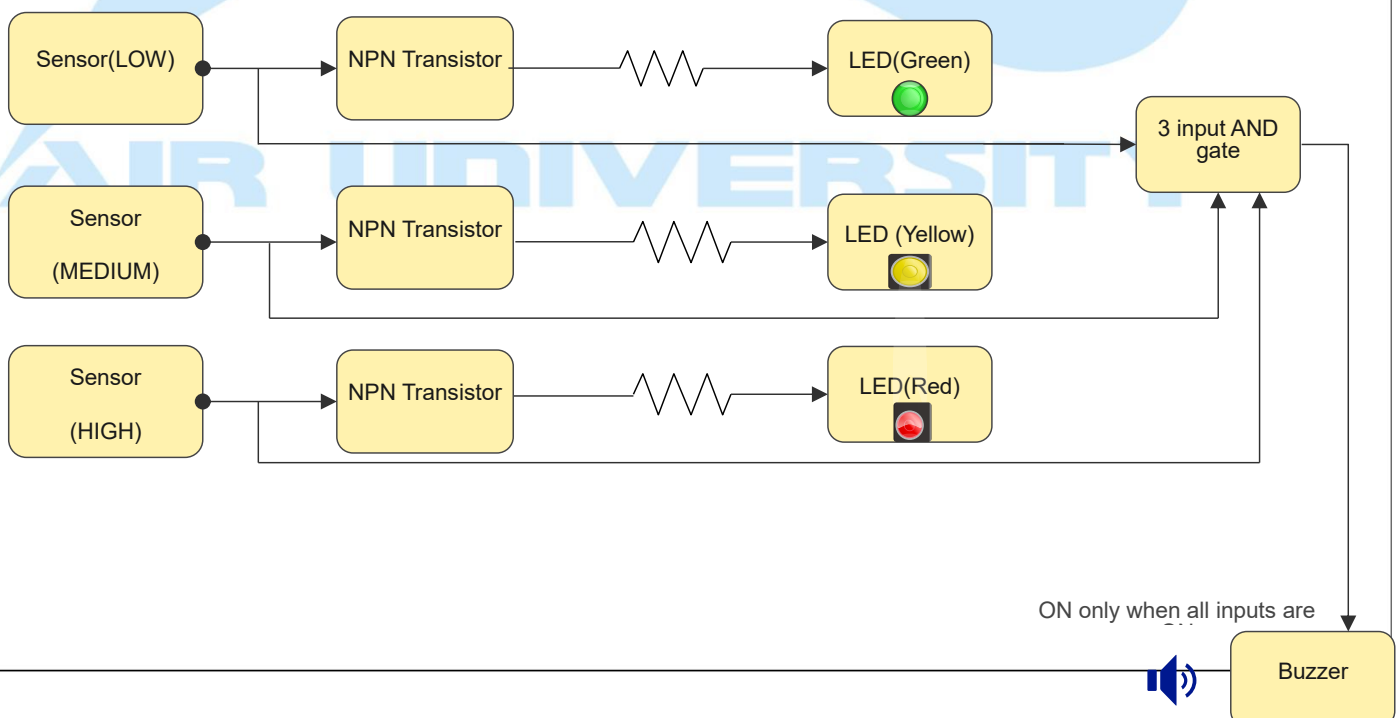
vi. Global Controls (Clock, Reset, Enabler)

Clock: Syncs the counter and the output flip-flops to move forward step-by-step.

Reset: Clears all stored memory back to zero.

Enabler: Must be active (1) for the output flip-flops to operate.

6. Block Diagram



i. Input Level Sensors

The system has three independent sensors representing different stages or thresholds:

- Sensor (LOW): Triggers first when the minimum water level is reached.
- Sensor (MEDIUM): Triggers when the middle water level is reached.
- Sensor (HIGH): Triggers when the maximum critical water level is reached.

ii. Driver & Visual Indicators (LEDs)

Each sensor has its own dedicated output channel:

NPN Transistor: Acts as an electronic switch. When a sensor outputs a high signal, it turns the transistor ON.

Resistor: Limits the electrical current to protect the LEDs from burning out.

LEDs (Green, Yellow, Red): Provide a visual status update.

- Low level = Green LED lights up.
- Medium level = Yellow LED lights up.
- High level = Red LED lights up.

iii. The Logic & Alarm System (AND Gate + Buzzer)

This is the master safety control mechanism of the circuit:

3-Input AND Gate: It acts as a logical decision-maker, monitoring the signal from all three sensors simultaneously.

The Logic Rule: An AND gate only outputs a high signal if all its inputs are active (1) at the same time.

Buzzer Activation: The Buzzer will remain completely silent during normal or partial levels. It will only turn ON when the system hits maximum capacity, meaning the Low, Medium, and High sensors are all triggered concurrently.

7. Truth table

A	B	C	x	y	z	u
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	1	0	0	0
1	0	1	1	0	0	0
1	1	0	1	1	0	0
1	1	1	1	1	1	1

Karnaugh maps

For u:

A\BC	00	01	11	10
0	0	0	0	0
1	0	0	①	0

$$u = ABC$$

For x:

A\BC	00	01	11	10
0	0	0	0	0
1	1	1	1	1

$$x = A$$

For y:

A\BC	00	01	11	10
0	0	0	0	0
1	0	0	1	1

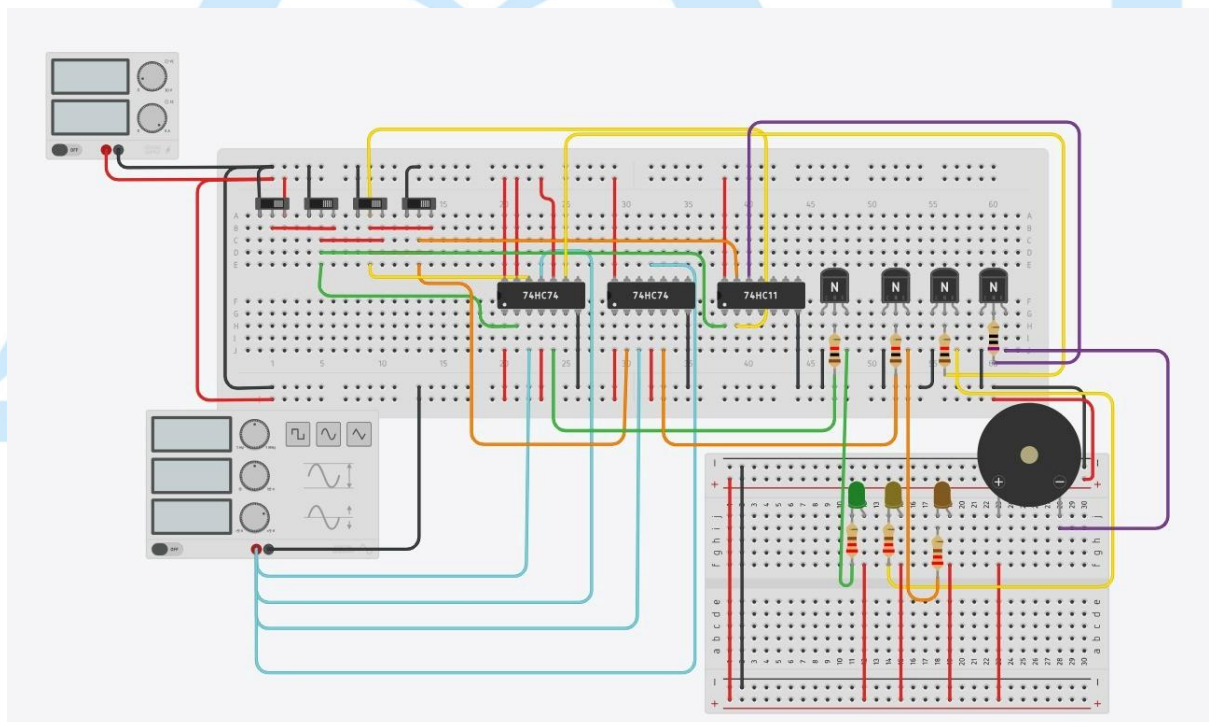
$$y = AB$$

For z:

A\BC	00	01	11	10
0	0	0	0	0
1	0	0	1	0

$$z = ABC$$

8. Tinker Cad



9. Implementation Steps

➤ Step 1: Workspace Setup

A new digital electronics simulation workspace opened in Tinkercad Circuits.

➤ Step 2: Component Selection

Dragged and dropped the required components from the Tinkercad library onto the workspace:

- 1x Large Breadboard & 1x Small Breadboard
- 2x 74HC74 (Dual D Flip-Flops with Preset and Clear)
- 1x 74HC11 (Triple 3-Input AND Gate)
- 4x Slide Switches (for manual inputs/control)
- 4x NPN Transistors (for switching/driving outputs)
- 3x LEDs (Green, Yellow, Red) and 1x Piezo Buzzer
- 1x DC Power Supply and 1x Function Generator (for clock signals)
- Resistors for current limiting.

➤ Step 3: Power Distribution (VCC and GND)

Connected the positive terminal of the DC Power Supply to the top power rail (Red) and the negative terminal to the ground rail (Black) of the main breadboard. Extended these power rails to the secondary small breadboard to ensure uniform 5V power distribution.

➤ Step 4: IC Placement and Powering

Placed the two 74HC74 ICs and one 74HC11 IC across the center dividing notch of the main breadboard. Wired Pin 14 of each IC to the 5V VCC rail and Pin 7 to the Ground rail.

➤ Step 5: Clock and Control Inputs Wiring

Connected the Function Generator to the clock pins of the D Flip-Flops (light blue wires) to supply synchronous sequential clock pulses. Wired the four Slide Switches at the top-left to provide static inputs/control logic to the sequential setup.

➤ Step 6: Combinational and Inter-flop Wiring

Connected the outputs Q and Q' of the D Flip-Flops back into the input stages and into the 74HC11 AND gate based on the circuit's state diagram logic using color-coded wires (Green, Orange, and Yellow) for clean troubleshooting.

➤ Step 7: Output Driver Stage (Transistors & Indicators)

Routed the final logic signals from the ICs to the bases of the four NPN Transistors through protective resistors. Connected the collectors/emitters to drive the Green, Yellow, and Brown/Red LEDs and the Piezo Buzzer on the secondary board.

➤ Step 8: Simulation and Verification

Clicked on "Start Simulation" in Tinkercad. Toggled the slide switches and monitored the LEDs and Buzzer states against the clock pulses to verify the truth table and state transitions.

10. Truth table and test cases

Current State (Q1 Q0)	Input (X)	Next State (Q1+ Q0+)	Outputs (LEDs / Buzzer)
00	0	01	Green LED ON
01	0	10	Yellow LED ON
10	0	11	Red LED ON
11	0	00	Buzzer Beeps (Alarm)

(b)

Test Case 1:

Initial State & First LED Activation

- **Condition:** The simulation is started with the DC Power Supply set exactly to 3.5V, and the clock/sequential logic triggers the first state.
- **Expected Output:** The Green LED should turn ON at the 3.5V operating voltage. All other LEDs and the buzzer should remain OFF.

➤ **Observed Output:** As soon as the simulation started, the Green LED illuminated successfully.

➤ **Status:** PASSED

Test Case 2:

Cumulative State Transition (LEDs Staying ON)

➤ **Condition:** The system receives subsequent clock pulses to move to the next logical states under 3.5V power.

➤ **Expected Output:** As the circuit transitions to the next states, the Next LED (Yellow) should turn ON, but the previous LED (Green) must remain ON. The lights should accumulate rather than turning off.

➤ **Observed Output:** The Yellow LED turned ON as expected, while the Green LED stayed completely illuminated. The circuit successfully held the previous states.

➤ **Status:** PASSED

Test Case 3:

All Switches ON Condition (Buzzer Trigger)

➤ **Condition:** All manual input control switches are toggled to the ON (HIGH) position simultaneously.

➤ **Expected Output:** The combinational logic (74HC11 AND gate) should decode that all inputs are HIGH, driving the NPN transistor to activate the Piezo Buzzer using the 3.5V supply rail.

➤ **Observed Output:** The moment all switches were moved to the ON position, the Piezo Buzzer triggered immediately, emitting sound/vibrations in Tinkercad.

➤ **Status:** PASSED

11. Result Analysis

The circuit was successfully designed and simulated on Tinkercad at an operating voltage of 3.5 V. The results analyze that the 74HC74 D-Flip Flops function effectively as a synchronous sequential system, moving through states upon receiving clock signals from the Function Generator. A key feature of this design is its cumulative output behavior: rather than turning off, previous LEDs remain illuminated as the next state triggers, creating a step-by-step visual progression (Green --> Green+ Yellow --> Green +Yellow+ Orange).

Furthermore, the combinational stage utilizing the 74HC11 (3-Input AND Gate) works as a strict safety or completion monitor. It successfully prevents the Piezo Buzzer from activating during intermediate states, triggering it only when the absolute condition of having all control switches set to HIGH is met. The NPN

transistors proved vital at 3.5V to properly drive the load current for the buzzer and individual LEDs."

12. Challenges and Solutions

Challenge 1: Low Voltage Operations (3.5V Constraint)

Problem: Standard TTL/CMOS ICs ideally prefer 5V. Operating the entire setup, especially multiple cumulative LEDs and a buzzer, at 3.5V risked weak logic levels and dim outputs.

Solution: NPN Transistors were utilized as electronic switches. Instead of forcing the IC pins to supply current directly at 3.5V, the ICs merely provided a small base current to switch the transistors, allowing the indicators to draw power efficiently from the main rail.

Challenge 2: Cumulative Logic Wiring

Problem: Designing the circuit so that previous LEDs stay ON required careful routing of the state outputs without creating short circuits or backward current feedback into the flip-flops.

Solution: Isolation of signals was achieved using correct node branching on the Tinkercad breadboard and verifying that the logic high state 1 remains latched until an explicit reset occurs.

13. Limitations:

Voltage Sensitivity: Since the circuit operates at a low threshold of 3.5V, any practical voltage drop across the breadboard rails on real hardware could cause the 74HC-series ICs to misbehave or freeze.

Rigid Logic Structure: The system is purely hardware-configured (hardwired). If the sequence order or the buzzer's activation condition needs to be changed, the entire physical wiring must be re-done.

14. Conclusion

In conclusion, this project successfully demonstrates the implementation of a specialized Sequential and Combinational Logic System operating under a specific 3.5V power constraint.

Through Tinkercad, we successfully verified a unique state machine where the outputs accumulate over time rather than resetting sequentially. The integration of the 74HC11 AND gate as a conditional trigger for the Piezo Buzzer highlights how basic logic gates can function as automated safety interlocks or condition checkers. Using transistors to bridge the low-voltage logic gates with higher-load actuators provided crucial hands-on insight into practical digital circuit design and optimization."